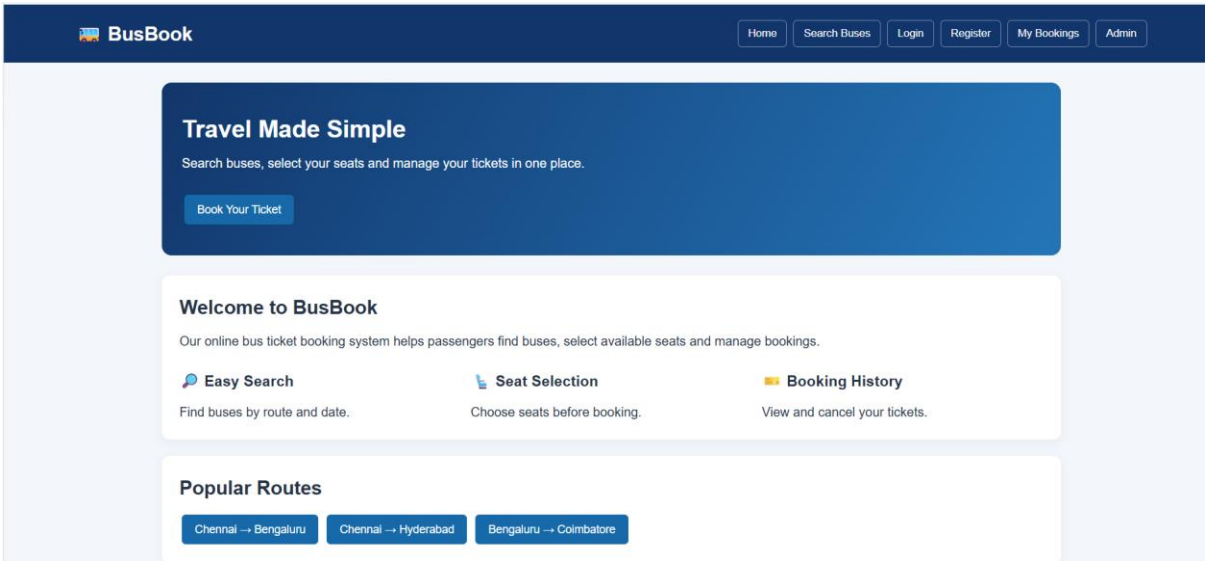
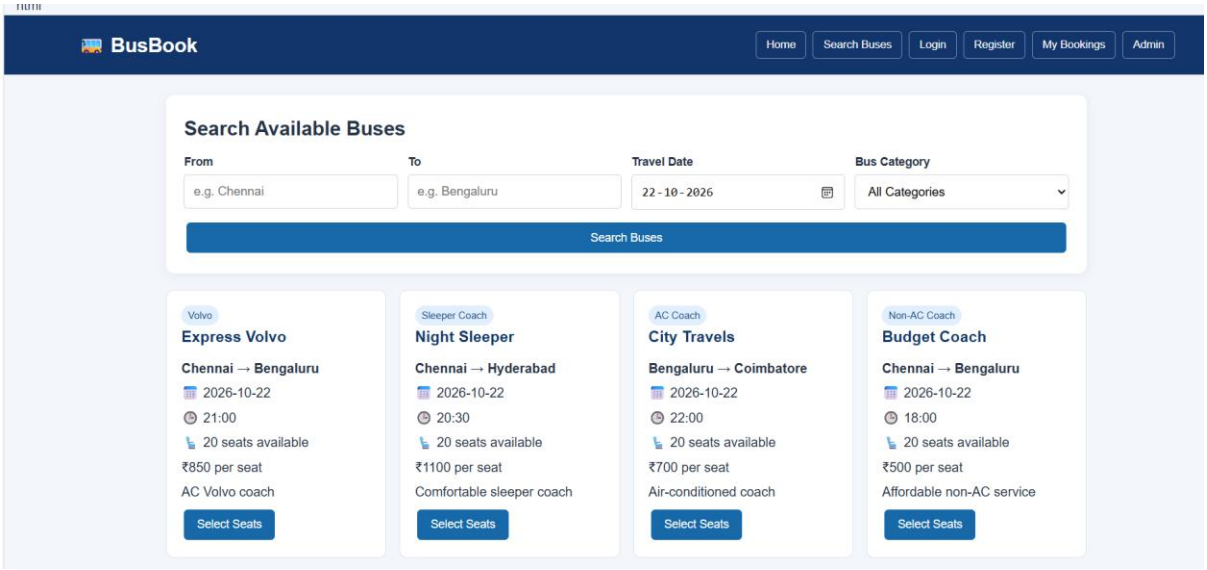
EXP 3-Online bus ticket booking





<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Online Bus Ticket Booking</title>

<style>

* {

```
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

body {
  font-family: Arial, sans-serif;
  background: #f3f6fb;
  color: #263247;
  line-height: 1.5;
}

header {
  background: #12366b;
  color: white;
  padding: 18px 6%;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 12px;
}

header h1 {
  font-size: 24px;
}

nav {
```

```
  display: flex;
  gap: 8px;
  flex-wrap: wrap;
}

nav button {
  background: transparent;
  color: white;
  border: 1px solid #ffffff70;
  padding: 8px 12px;
  border-radius: 5px;
  cursor: pointer;
}

nav button:hover {
  background: #28569a;
}

main {
  max-width: 1150px;
  margin: 25px auto;
  padding: 0 16px;
}

.page {
  display: none;
}
```

```
.page.active {
  display: block;
}

.hero {
  background: linear-gradient(120deg, #12366b, #2476b9);
  color: white;
  border-radius: 12px;
  padding: 35px 25px;
  margin-bottom: 25px;
}

.hero h2 {
  font-size: 30px;
  margin-bottom: 8px;
}

.card {
  background: white;
  border-radius: 10px;
  padding: 22px;
  margin-bottom: 20px;
  box-shadow: 0 3px 12px #1c355410;
}

h2, h3 {
  margin-bottom: 12px;
}
```

```
form {
  display: grid;
  gap: 12px;
}

.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
  gap: 12px;
}

label {
  font-weight: bold;
  font-size: 14px;
}

input, select, textarea {
  width: 100%;
  padding: 11px;
  margin-top: 5px;
  border: 1px solid #cbd5e1;
  border-radius: 5px;
  font-size: 15px;
  background: white;
}

textarea {
```

```
    min-height: 85px;
    resize: vertical;
}

.btn {
    background: #1769aa;
    color: white;
    padding: 10px 16px;
    border: none;
    border-radius: 5px;
    cursor: pointer;
    font-size: 14px;
    margin: 3px;
}

.btn:hover {
    background: #104c80;
}

.btn-danger {
    background: #c0392b;
}

.btn-secondary {
    background: #64748b;
}

.btn-success {
```

```
  background: #198754;
}

.bus-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
  gap: 16px;
}

.bus-card {
  border: 1px solid #dce4ef;
  padding: 18px;
  border-radius: 9px;
  background: white;
}

.bus-card h3 {
  color: #12366b;
}

.bus-card p {
  margin: 6px 0;
}

.badge {
  display: inline-block;
  background: #e2efff;
  color: #174c87;
```

```
    padding: 3px 8px;
    border-radius: 20px;
    font-size: 12px;
}

.seats {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 9px;
    max-width: 350px;
    margin: 15px 0;
}

.seat {
    padding: 10px 4px;
    border: 1px solid #b8c6d8;
    background: #edf2f7;
    border-radius: 5px;
    cursor: pointer;
}

.seat.selected {
    background: #198754;
    color: white;
}

.seat.booked {
    background: #cbd5e1;
```

```
    color: #64748b;
    cursor: not-allowed;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 12px;
    background: white;
}

th, td {
    padding: 10px;
    border: 1px solid #dce4ef;
    text-align: left;
}

th {
    background: #eaf1f9;
}

.table-wrap {
    overflow-x: auto;
}

.small-text {
    color: #64748b;
    font-size: 13px;
```

```
}

#message {
  display: none;
  background: #d9f7e5;
  color: #155b34;
  border: 1px solid #9eddb5;
  border-radius: 6px;
  padding: 12px;
  margin-bottom: 15px;
}

footer {
  background: #12366b;
  color: white;
  text-align: center;
  padding: 20px;
  margin-top: 35px;
}

@media (max-width: 600px) {
  header {
    align-items: flex-start;
  }

  header h1 {
    font-size: 21px;
  }
```

```
    nav button {
      padding: 7px;
      font-size: 12px;
    }

    .hero h2 {
      font-size: 24px;
    }

    .card {
      padding: 15px;
    }

    th, td {
      padding: 7px;
      font-size: 12px;
    }
  }
</style>
</head>

<body>

<header>
  <h1>🚌 BusBook</h1>
  <nav>
    <button onclick="showPage('home')">Home</button>
```

```
      <button onclick="showPage('search')">Search Buses</button>
      <button onclick="showPage('login')">Login</button>
      <button onclick="showPage('register')">Register</button>
      <button onclick="showPage('dashboard')">My Bookings</button>
      <button onclick="showPage('admin')">Admin</button>
    </nav>
</header>

<main>
<div id="message"></div>

<!-- HOME -->
<section id="home" class="page active">
  <div class="hero">
    <h2>Travel Made Simple</h2>
    <p>Search buses, select your seats and manage your tickets in one place.</p>
    <br>
    <button class="btn" onclick="showPage('search')">Book Your Ticket</button>
  </div>

  <div class="card">
    <h2>Welcome to BusBook</h2>
    <p>Our online bus ticket booking system helps passengers find buses, select available seats and manage bookings.</p>
    <br>
    <div class="form-grid">
      <div><h3> 🔎 Easy Search</h3><p>Find buses by route and date.</p></div>
      <div><h3> 💺 Seat Selection</h3><p>Choose seats before booking.</p></div>
```

```
        <div><h3>🎫 Booking History</h3><p>View and cancel your tickets.</p></div>
      </div>
    </div>

    <div class="card">
      <h2>Popular Routes</h2>
      <button class="btn" onclick="quickSearch('Chennai','Bengaluru')">Chennai → Bengaluru</button>
      <button class="btn" onclick="quickSearch('Chennai','Hyderabad')">Chennai → Hyderabad</button>
      <button class="btn" onclick="quickSearch('Bengaluru','Coimbatore')">Bengaluru → Coimbatore</button>
    </div>
  </section>

  <!-- SEARCH -->
  <section id="search" class="page">
    <div class="card">
      <h2>Search Available Buses</h2>
      <form onsubmit="searchBuses(event)">
        <div class="form-grid">
          <label>From
            <input id="source" required placeholder="e.g. Chennai">
          </label>
          <label>To
            <input id="destination" required placeholder="e.g. Bengaluru">
          </label>
          <label>Travel Date
            <input id="travelDate" type="date" required>
```

```
          </label>
          <label>Bus Category
            <select id="categoryFilter">
              <option value="">All Categories</option>
              <option>Volvo</option>
              <option>Sleeper Coach</option>
              <option>AC Coach</option>
              <option>Non-AC Coach</option>
            </select>
          </label>
        </div>
        <button class="btn" type="submit">Search Buses</button>
      </form>
    </div>

    <div id="busResults" class="bus-grid"></div>
</section>

<!-- SEAT BOOKING -->
<section id="booking" class="page">
    <div class="card">
      <h2>Book Your Ticket</h2>
      <div id="selectedBusDetails"></div>

      <h3>Select Your Seat</h3>
      <p class="small-text">Green indicates your selected seat. Grey indicates a booked seat.</p>
      <div id="seatLayout" class="seats"></div>
```

```
<p id="seatInfo">No seat selected.</p>


<hr>
<br>
<h3>Passenger Details</h3>
<form onsubmit="confirmBooking(event)">
  <div class="form-grid">
    <label>Passenger Name
      <input id="passengerName" required>
    </label>
    <label>Age
      <input id="passengerAge" type="number" min="1" max="120" required>
    </label>
    <label>Gender
      <select id="passengerGender" required>
        <option value="">Select</option>
        <option>Male</option>
        <option>Female</option>
        <option>Other</option>
      </select>
    </label>
    <label>Contact Email
      <input id="passengerEmail" type="email" required>
    </label>
  </div>

  <h3>Payment Method (Demo)</h3>
  <select id="paymentMethod" required>
```

```
            <option value="">Choose payment method</option>
            <option>UPI (Demo)</option>
            <option>Debit/Credit Card (Demo)</option>
            <option>Net Banking (Demo)</option>
          </select>
          <p class="small-text">This lab demonstration does not process real payments.</p>
          <button type="submit" class="btn btn-success">Confirm Booking</button>
          <button type="button" class="btn btn-secondary" onclick="showPage('search')">Back to Search</button>
        </form>
      </div>
    </section>

    <!-- REGISTER -->
    <section id="register" class="page">
      <div class="card">
        <h2>Create an Account</h2>
        <form onsubmit="registerUser(event)">
          <label>Full Name
            <input id="regName" required>
          </label>
          <label>Email
            <input id="regEmail" type="email" required>
          </label>
          <label>Password
            <input id="regPassword" type="password" minlength="4" required>
          </label>
          <label>Confirm Password
```

```
          <input id="regConfirm" type="password" minlength="4" required>
        </label>
        <button class="btn" type="submit">Register</button>
      </form>
      <p>Already registered? <a href="#" onclick="showPage('login')">Login here</a></p>
    </div>
</section>

<!-- LOGIN -->
<section id="login" class="page">
    <div class="card">
      <h2>User Login</h2>
      <form onsubmit="loginUser(event)">
        <label>Email
          <input id="loginEmail" type="email" required>
        </label>
        <label>Password
          <input id="loginPassword" type="password" required>
        </label>
        <button class="btn" type="submit">Login</button>
        <button class="btn btn-secondary" type="button"
onclick="forgotPassword()">Forgot Password?</button>
      </form>
      <br>
      <p class="small-text">Demo user: user@bus.com / user123</p>
      <p class="small-text">Demo admin: admin@bus.com / admin123</p>
    </div>
</section>
```

```
<!-- USER DASHBOARD -->
<section id="dashboard" class="page">
  <div class="card">
    <h2>My Dashboard</h2>
    <p id="userWelcome">Please log in to view your bookings.</p>
    <button class="btn" onclick="showPage('search')">Book Another Ticket</button>
    <button class="btn btn-secondary" onclick="changePassword()">Change Password</button>
    <button class="btn btn-danger" onclick="logout()">Logout</button>
  </div>

  <div class="card">
    <h3>My Booking History</h3>
    <div id="myBookings" class="table-wrap"></div>
  </div>

  <div class="card">
    <h3>Post a Comment or Review</h3>
    <form onsubmit="postComment(event)">
      <label>Select Bus
        <select id="commentBus"></select>
      </label>
      <label>Your Comment
        <textarea id="commentText" required placeholder="Share your travel experience"></textarea>
      </label>
      <button class="btn" type="submit">Post Comment</button>
    </form>
```

```
      <div id="commentList"></div>
    </div>
</section>

<!-- ADMIN PANEL -->
<section id="admin" class="page">
    <div class="card">
      <h2>Admin Panel</h2>
      <p id="adminStatus">Log in with the demo admin account to manage the system.</p>
      <button class="btn" onclick="showPage('login')">Admin Login</button>
    </div>

    <div id="adminTools" style="display:none">
      <div class="card">
        <h3>Add or Update Bus Information</h3>
        <form onsubmit="saveBus(event)">
          <input type="hidden" id="editBusId">
          <div class="form-grid">
            <label>Bus Name
              <input id="busName" required>
            </label>
            <label>Bus Category
              <select id="busCategory" required>
                <option>Volvo</option>
                <option>Sleeper Coach</option>
                <option>AC Coach</option>
                <option>Non-AC Coach</option>
```

```
            </select>
          </label>
          <label>Source
            <input id="busSource" required>
          </label>
          <label>Destination
            <input id="busDestination" required>
          </label>
          <label>Departure Time
            <input id="busTime" type="time" required>
          </label>
          <label>Ticket Price (₹)
            <input id="busPrice" type="number" min="1" required>
          </label>
          <label>Total Seats
            <input id="busSeats" type="number" min="1" max="40" required>
          </label>
          <label>Travel Date
            <input id="busDate" type="date" required>
          </label>
          <label>Bus Description
            <input id="busDescription" placeholder="AC sleeper, charging ports...">
          </label>
        </div>
        <button class="btn btn-success" type="submit">Save Bus</button>
        <button class="btn btn-secondary" type="button"
onclick="clearBusForm()">Clear Form</button>
      </form>
```

```
      </div>

      <div class="card">
        <h3>Manage Buses</h3>
        <div id="allBuses" class="table-wrap"></div>
      </div>

      <div class="card">
        <h3>Registered Users</h3>
        <div id="allUsers" class="table-wrap"></div>
      </div>

      <div class="card">
        <h3>Booking Reports</h3>
        <button class="btn" onclick="generateReport()">Generate Report</button>
        <button class="btn btn-secondary" onclick="downloadReport()">Download CSV
Report</button>
        <div id="reportOutput"></div>
      </div>

      <div class="card">
        <h3>Manage Comments</h3>
        <div id="adminComments"></div>
      </div>
    </div>
</section>

<!-- ABOUT -->
```

```
<section id="about" class="page">
  <div class="card">
    <h2>About BusBook</h2>
    <p>BusBook is a demonstration website designed to simplify bus searching, seat selection and ticket booking. It includes passenger and administrator interfaces for an Internet Programming lab experiment.</p>
  </div>
</section>

<!-- CONTACT -->
<section id="contact" class="page">
  <div class="card">
    <h2>Contact Us</h2>
    <form onsubmit="contactUs(event)">
      <label>Name <input required id="contactName"></label>
      <label>Email <input type="email" required id="contactEmail"></label>
      <label>Message <textarea required id="contactMessage"></textarea></label>
      <button class="btn" type="submit">Send Message</button>
    </form>
  </div>
</section>

</main>

<footer>
  <p>© 2026 BusBook | Internet Programming Lab Experiment</p>
  <p>Home · Bus Search · Booking · Contact</p>
</footer>
```

```
<script>

let users = [

  {name: "Demo User", email: "user@bus.com", password: "user123", role: "user"},

  {name: "System Admin", email: "admin@bus.com", password: "admin123", role: "admin"}

];


let buses = [

  {id: 1, name: "Express Volvo", category: "Volvo", source: "Chennai", destination: "Bengaluru", time: "21:00", price: 850, seats: 20, date: "2026-12-20", description: "AC Volvo coach"},

  {id: 2, name: "Night Sleeper", category: "Sleeper Coach", source: "Chennai", destination: "Hyderabad", time: "20:30", price: 1100, seats: 20, date: "2026-12-20", description: "Comfortable sleeper coach"},

  {id: 3, name: "City Travels", category: "AC Coach", source: "Bengaluru", destination: "Coimbatore", time: "22:00", price: 700, seats: 20, date: "2026-12-20", description: "Air-conditioned coach"},

  {id: 4, name: "Budget Coach", category: "Non-AC Coach", source: "Chennai", destination: "Bengaluru", time: "18:00", price: 500, seats: 20, date: "2026-12-20", description: "Affordable non-AC service"}

];


let bookings = [];

let comments = [];

let currentUser = null;

let selectedBusId = null;

let selectedSeat = null;

let nextUserId = 3;

let nextBusId = 5;

let nextBookingId = 1001;
```

```
function showMessage(text) {
  const box = document.getElementById("message");
  box.textContent = text;
  box.style.display = "block";
  window.scrollTo({top: 0, behavior: "smooth"});
}

function showPage(id) {
  document.querySelectorAll(".page").forEach(p => p.classList.remove("active"));
  document.getElementById(id).classList.add("active");
  document.getElementById("message").style.display = "none";

  if (id === "dashboard") renderDashboard();
  if (id === "admin") renderAdmin();
  if (id === "search") renderBuses(buses);
  if (id === "dashboard") renderComments();
  window.scrollTo(0, 0);
}

function quickSearch(from, to) {
  document.getElementById("source").value = from;
  document.getElementById("destination").value = to;
  showPage("search");
}

function searchBuses(event) {
  event.preventDefault();
```

```
  const from = document.getElementById("source").value.trim().toLowerCase();
  const to = document.getElementById("destination").value.trim().toLowerCase();
  const date = document.getElementById("travelDate").value;
  const category = document.getElementById("categoryFilter").value;

  const results = buses.filter(bus =>
    bus.source.toLowerCase() === from &&
    bus.destination.toLowerCase() === to &&
    bus.date === date &&
    (!category || bus.category === category)
  );

  renderBuses(results);
  if (results.length === 0) {
    document.getElementById("busResults").innerHTML =
      "<div class='card'>No buses found for this route and date. Try another date or route.</div>";
  }
}

function renderBuses(list) {
  const area = document.getElementById("busResults");

  if (!list.length) {
    area.innerHTML = "<div class='card'>No buses currently listed. Search another route or ask the admin to add a bus.</div>";
    return;
  }
```

```
  area.innerHTML = list.map(bus => {
    const bookedCount = bookings.filter(b => b.busId === bus.id && b.status === "Booked").length;
    const available = Math.max(0, bus.seats - bookedCount);

    return `
    <div class="bus-card">
      <span class="badge">${bus.category}</span>
      <h3>${escapeHTML(bus.name)}</h3>
      <p><b>${escapeHTML(bus.source)} → ${escapeHTML(bus.destination)}</b></p>
      <p>🗓️ ${bus.date}</p>
      <p>🕒 ${bus.time}</p>
      <p>💺 ${available} seats available</p>
      <p>₹${bus.price} per seat</p>
      <p>${escapeHTML(bus.description || "")}</p>
      <button class="btn" ${available === 0 ? "disabled" : ""} onclick="startBooking(${bus.id})">
        ${available === 0 ? "Sold Out" : "Select Seats"}
      </button>
    </div>`;
  }).join("");
}

function startBooking(busId) {
  if (!currentUser) {
    showMessage("Please log in before booking a ticket.");
    showPage("login");
    return;
```

```
  }

  const bus = buses.find(b => b.id === busId);
  if (!bus) return;

  selectedBusId = busId;
  selectedSeat = null;

  document.getElementById("selectedBusDetails").innerHTML =
    `<h3>${escapeHTML(bus.name)}</h3>
    <p>${escapeHTML(bus.source)} → ${escapeHTML(bus.destination)}</p>
    <p>Date: ${bus.date} | Departure: ${bus.time}</p>
    <p>Price: ₹${bus.price}</p>`;

  document.getElementById("passengerName").value = currentUser.name;
  document.getElementById("passengerEmail").value = currentUser.email;

  renderSeats();
  showPage("booking");
}

function renderSeats() {
  const bus = buses.find(b => b.id === selectedBusId);
  if (!bus) return;

  const bookedSeats = bookings
    .filter(b => b.busId === bus.id && b.status === "Booked")
    .map(b => b.seat);
```

```
  let html = "";
  for (let i = 1; i <= bus.seats; i++) {
    const seat = "S" + i;
    const isBooked = bookedSeats.includes(seat);

    html += `<button type="button"
      class="seat ${isBooked ? "booked" : ""} ${selectedSeat === seat ? "selected" : ""}"
      ${isBooked ? "disabled" : ""}
      onclick="chooseSeat('${seat}')">${seat}</button>`;
  }

  document.getElementById("seatLayout").innerHTML = html;
  document.getElementById("seatInfo").textContent =
    selectedSeat ? "Selected seat: " + selectedSeat : "No seat selected.";
}

function chooseSeat(seat) {
  selectedSeat = seat;
  renderSeats();
}

function confirmBooking(event) {
  event.preventDefault();

  if (!currentUser) {
    showMessage("Please log in first.");
    showPage("login");
```

```
    return;
}

if (!selectedSeat) {
    alert("Please select a seat before confirming.");
    return;
}

const bus = buses.find(b => b.id === selectedBusId);
if (!bus) return;

const seatTaken = bookings.some(b =>
    b.busId === bus.id && b.seat === selectedSeat && b.status === "Booked"
);

if (seatTaken) {
    alert("This seat has already been booked. Please select another seat.");
    selectedSeat = null;
    renderSeats();
    return;
}

const booking = {
    id: nextBookingId++,
    userEmail: currentUser.email,
    busId: bus.id,
    busName: bus.name,
    source: bus.source,
```

```
      destination: bus.destination,
      date: bus.date,
      time: bus.time,
      seat: selectedSeat,
      passengerName: document.getElementById("passengerName").value.trim(),
      age: document.getElementById("passengerAge").value,
      gender: document.getElementById("passengerGender").value,
      email: document.getElementById("passengerEmail").value.trim(),
      payment: document.getElementById("paymentMethod").value,
      amount: bus.price,
      status: "Booked"
    };

    bookings.push(booking);

    alert("Booking confirmed!\nBooking ID: " + booking.id +
        "\nSeat: " + booking.seat +
        "\nAmount: ₹" + booking.amount +
        "\n\nThis is a demonstration booking; no real payment or email was sent.");

    selectedSeat = null;
    showPage("dashboard");
    showMessage("Ticket booked successfully. Booking ID: " + booking.id);
}

function registerUser(event) {
    event.preventDefault();
```

```
  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;
  const confirm = document.getElementById("regConfirm").value;

  if (password !== confirm) {
    alert("Passwords do not match.");
    return;
  }

  if (users.some(u => u.email.toLowerCase() === email)) {
    alert("An account with this email already exists.");
    return;
  }

  users.push({id: nextUserId++, name, email, password, role: "user"});
  alert("Registration successful. Please log in.");
  event.target.reset();
  showPage("login");
}

function loginUser(event) {
  event.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u =>
```

```
    u.email.toLowerCase() === email && u.password === password
  );

  if (!user) {
    alert("Invalid email or password.");
    return;
  }

  currentUser = user;
  alert("Welcome, " + user.name + "!");

  if (user.role === "admin") {
    showPage("admin");
  } else {
    showPage("dashboard");
  }
}

function logout() {
  currentUser = null;
  showMessage("You have logged out.");
  showPage("home");
}

function forgotPassword() {
  const email = prompt("Enter your registered email address:");
  if (!email) return;
```

```
  const user = users.find(u => u.email.toLowerCase() === email.trim().toLowerCase());

  if (!user) {
    alert("No account found with that email.");
    return;
  }

  const newPassword = prompt("Demo password reset: enter a new password (at least 4 characters):");
  if (!newPassword || newPassword.length < 4) {
    alert("Password must contain at least 4 characters.");
    return;
  }

  user.password = newPassword;
  alert("Password changed for this demo session. No reset email was sent.");
}

function changePassword() {
  if (!currentUser) {
    alert("Please log in first.");
    return;
  }

  const oldPassword = prompt("Enter current password:");
  if (oldPassword !== currentUser.password) {
    alert("Current password is incorrect.");
    return;
```

```
  }

  const newPassword = prompt("Enter new password (at least 4 characters):");
  if (!newPassword || newPassword.length < 4) {
    alert("Password must contain at least 4 characters.");
    return;
  }

  currentUser.password = newPassword;
  alert("Password changed successfully for this session.");
}

function renderDashboard() {
  const welcome = document.getElementById("userWelcome");

  if (!currentUser) {
    welcome.textContent = "Please log in to view your bookings.";
    document.getElementById("myBookings").innerHTML =
      "<p>Log in to see your booking history.</p>";
    return;
  }

  welcome.textContent = "Welcome, " + currentUser.name + " (" + currentUser.email +
")";

  const mine = bookings.filter(b => b.userEmail === currentUser.email);

  if (!mine.length) {
```

```
    document.getElementById("myBookings").innerHTML =
      "<p>You have not booked any tickets yet.</p>";
    return;
  }

  document.getElementById("myBookings").innerHTML = `
  <table>
    <tr><th>ID</th><th>Route</th><th>Date</th><th>Seat</th><th>Amount</th><th>Status</th><th>Action</th></tr>
    ${mine.map(b => `<tr>
      <td>${b.id}</td>
      <td>${escapeHTML(b.source)} → ${escapeHTML(b.destination)}</td>
      <td>${b.date}</td>
      <td>${b.seat}</td>
      <td>₹${b.amount}</td>
      <td>${b.status}</td>
      <td>${b.status === "Booked"
        ? `<button class="btn btn-danger" onclick="cancelBooking(${b.id})">Cancel</button>`
        : "—"}</td>
    </tr>`).join("")}
  </table>`;
}

function cancelBooking(id) {
  const booking = bookings.find(b => b.id === id);

  if (!currentUser || !booking || booking.userEmail !== currentUser.email) {
    alert("You can only cancel your own bookings.");
```

```
    return;
  }

  if (booking.status !== "Booked") {
    alert("This booking is already cancelled.");
    return;
  }

  if (confirm("Cancel booking #" + id + "?")) {
    booking.status = "Cancelled";
    renderDashboard();
    showMessage("Booking cancelled. The seat is now available again.");
  }
}

function saveBus(event) {
  event.preventDefault();

  if (!isAdmin()) {
    alert("Please log in as admin first.");
    return;
  }

  const id = Number(document.getElementById("editBusId").value);
  const data = {
    name: document.getElementById("busName").value.trim(),
    category: document.getElementById("busCategory").value,
    source: document.getElementById("busSource").value.trim(),
```

```
    destination: document.getElementById("busDestination").value.trim(),
    time: document.getElementById("busTime").value,
    price: Number(document.getElementById("busPrice").value),
    seats: Number(document.getElementById("busSeats").value),
    date: document.getElementById("busDate").value,
    description: document.getElementById("busDescription").value.trim()
  };

  if (data.source.toLowerCase() === data.destination.toLowerCase()) {
    alert("Source and destination must be different.");
    return;
  }

  if (id) {
    const bus = buses.find(b => b.id === id);
    if (!bus) return;

    const bookedCount = bookings.filter(b => b.busId === id && b.status ===
"Booked").length;
    if (data.seats < bookedCount) {
      alert("Total seats cannot be fewer than the number of active bookings.");
      return;
    }

    Object.assign(bus, data);
    alert("Bus information updated.");
  } else {
    buses.push({id: nextBusId++, ...data});
```

```
      alert("New bus added.");
    }

    clearBusForm();
    renderAdmin();
    renderBuses(buses);
}

function editBus(id) {
    if (!isAdmin()) return;

    const bus = buses.find(b => b.id === id);
    if (!bus) return;

    document.getElementById("editBusId").value = bus.id;
    document.getElementById("busName").value = bus.name;
    document.getElementById("busCategory").value = bus.category;
    document.getElementById("busSource").value = bus.source;
    document.getElementById("busDestination").value = bus.destination;
    document.getElementById("busTime").value = bus.time;
    document.getElementById("busPrice").value = bus.price;
    document.getElementById("busSeats").value = bus.seats;
    document.getElementById("busDate").value = bus.date;
    document.getElementById("busDescription").value = bus.description || "";

    window.scrollTo(0, 0);
}
```

```
function deleteBus(id) {
  if (!isAdmin()) return;

  const hasActiveBookings = bookings.some(b => b.busId === id && b.status === "Booked");

  if (hasActiveBookings) {
    alert("Cancel active bookings before deleting this bus.");
    return;
  }

  if (confirm("Delete this bus?")) {
    buses = buses.filter(b => b.id !== id);
    renderAdmin();
  }
}

function clearBusForm() {
  document.getElementById("editBusId").value = "";
  document.querySelector("#adminTools form").reset();
}

function isAdmin() {
  return currentUser && currentUser.role === "admin";
}

function renderAdmin() {
  const tools = document.getElementById("adminTools");
```

```
  const status = document.getElementById("adminStatus");


  if (!isAdmin()) {
    tools.style.display = "none";
    status.textContent = "Log in with admin@bus.com / admin123 to manage the system.";
    return;
  }


  tools.style.display = "block";
  status.textContent = "Logged in as administrator: " + currentUser.email;


  document.getElementById("allBuses").innerHTML = `
  <table>
    <tr><th>ID</th><th>Bus</th><th>Route</th><th>Date</th><th>Price</th><th>Seats</th><th>Actions</th></tr>
    ${buses.map(b => `<tr>
      <td>${b.id}</td>
      <td>${escapeHTML(b.name)}<br>${escapeHTML(b.category)}</td>
      <td>${escapeHTML(b.source)} → ${escapeHTML(b.destination)}</td>
      <td>${b.date}<br>${b.time}</td>
      <td>₹${b.price}</td>
      <td>${b.seats}</td>
      <td>
        <button class="btn" onclick="editBus(${b.id})">Edit</button>
        <button class="btn btn-danger" onclick="deleteBus(${b.id})">Delete</button>
      </td>
    </tr>`).join("")}
  </table>`;
```

```
  document.getElementById("allUsers").innerHTML = `
  <table>
    <tr><th>Name</th><th>Email</th><th>Role</th><th>Action</th></tr>
    ${users.map(u => `<tr>
      <td>${escapeHTML(u.name)}</td>
      <td>${escapeHTML(u.email)}</td>
      <td>${u.role}</td>
      <td>${u.role === "user"
        ? `<button class="btn btn-danger"
onclick="removeUser('${escapeHTML(u.email)}')">Remove</button>`
        : "Protected"}</td>
    </tr>`).join("")}
  </table>`;

  document.getElementById("commentBus").innerHTML =
    buses.map(b => `<option value="${b.id}">${escapeHTML(b.name)} —
${escapeHTML(b.source)} to ${escapeHTML(b.destination)}</option>`).join("");

  renderAdminComments();
}

function removeUser(email) {
  if (!isAdmin()) return;

  if (bookings.some(b => b.userEmail === email && b.status === "Booked")) {
    alert("This user has active bookings. Cancel them before removing the user.");
    return;
  }
```

```
  if (confirm("Remove user " + email + "?")) {
    users = users.filter(u => u.email !== email);
    renderAdmin();
  }
}

function generateReport() {
  if (!isAdmin()) return;

  const active = bookings.filter(b => b.status === "Booked");
  const cancelled = bookings.filter(b => b.status === "Cancelled");
  const revenue = active.reduce((sum, b) => sum + b.amount, 0);

  document.getElementById("reportOutput").innerHTML = `
    <h3>Booking Report</h3>
    <p>Total buses: ${buses.length}</p>
    <p>Registered users: ${users.length}</p>
    <p>Total bookings: ${bookings.length}</p>
    <p>Active bookings: ${active.length}</p>
    <p>Cancelled bookings: ${cancelled.length}</p>
    <p>Value of active bookings: ₹${revenue}</p>`;
}

function downloadReport() {
  if (!isAdmin()) {
    alert("Admin login required.");
    return;
```

```
  }

  const rows = [
    ["Booking ID", "Passenger", "Email", "Bus", "Source", "Destination", "Date", "Seat",
"Amount", "Status"],
    ...bookings.map(b => [
      b.id, b.passengerName, b.email, b.busName, b.source,
      b.destination, b.date, b.seat, b.amount, b.status
    ])
  ];

  const csv = rows.map(row => row.map(value =>
    '"' + String(value).replace(/"/g, '""') + '"'
  ).join(",")).join("\n");

  const blob = new Blob([csv], {type: "text/csv;charset=utf-8;"});
  const url = URL.createObjectURL(blob);
  const link = document.createElement("a");
  link.href = url;
  link.download = "bus-booking-report.csv";
  link.click();
  URL.revokeObjectURL(url);
}

function postComment(event) {
  event.preventDefault();

  if (!currentUser || currentUser.role !== "user") {
```

```
    alert("Please log in as a user to post a comment.");
    return;
  }

  const busId = Number(document.getElementById("commentBus").value);
  const text = document.getElementById("commentText").value.trim();

  if (!text) return;

  comments.push({
    id: Date.now(),
    user: currentUser.name,
    busId,
    text
  });

  document.getElementById("commentText").value = "";
  renderComments();
  alert("Comment posted.");
}

function renderComments() {
  const area = document.getElementById("commentList");
  if (!area) return;

  if (!comments.length) {
    area.innerHTML = "<p>No comments yet.</p>";
    return;
```

```
  }

  area.innerHTML = comments.map(c => {
    const bus = buses.find(b => b.id === c.busId);
    return `<div class="card">
      <b>${escapeHTML(c.user)}</b>
      <p>${escapeHTML(bus ? bus.name : "Bus")}</p>
      <p>${escapeHTML(c.text)}</p>
    </div>`;
  }).join("");
}

function renderAdminComments() {
  const area = document.getElementById("adminComments");
  if (!area) return;

  if (!comments.length) {
    area.innerHTML = "<p>No comments have been posted.</p>";
    return;
  }

  area.innerHTML = comments.map(c => `
    <div class="card">
      <b>${escapeHTML(c.user)}</b>
      <p>${escapeHTML(c.text)}</p>
      <button class="btn btn-danger" onclick="deleteComment(${c.id})">Delete
Comment</button>
    </div>`).join("");
```

```
}

function deleteComment(id) {
  if (!isAdmin()) return;

  comments = comments.filter(c => c.id !== id);
  renderAdminComments();
  renderComments();
}

function contactUs(event) {
  event.preventDefault();
  alert("Thank you for contacting us. This demo does not send messages to a server.");
  event.target.reset();
}

function escapeHTML(value) {
  return String(value ?? "").replace(/[&<>"']/g, char => ({
    "&": "&amp;",
    "<": "&lt;",
    ">": "&gt;",
    '"': "&quot;",
    "'": "&#39;"
  }[char]));
}

// Use today's date as the default search date.
const today = new Date();
```

```
const localToday = [
  today.getFullYear(),
  String(today.getMonth() + 1).padStart(2, "0"),
  String(today.getDate()).padStart(2, "0")
].join("-");

document.getElementById("travelDate").value = localToday;
document.getElementById("travelDate").min = localToday;
document.getElementById("busDate").value = localToday;
document.getElementById("busDate").min = localToday;

// Keep the example buses searchable from today's date.
buses.forEach(bus => bus.date = localToday);

renderBuses(buses);
</script>

</body>
</html>
```